

ReefSafe

Evidence-led reef screening for sustainable tourism management

Malaysia Data Innovation Talent × DOSM Datathon 2026

Theme: Machine Learning and Artificial Intelligence for Sustainable Tourism

Submission report

Executive summary

ReefSafe is a screening system for marine-park teams. It combines long-running reef observations with NOAA thermal-stress records to identify islands that merit earlier field checks. It does not calculate legal visitor caps, estimate tourism revenue at island level, or prove that tourism caused a change in coral cover.

Evidence base	Coverage	Decision output
404 observed island-years	56 islands, 2012–2025	Ranked field-check priority
Past-only model evaluation	183 held-forward observations	Prediction band + uncertainty
Observed stressor flags	Thermal, anchor, waste/pollution	Evidence-matched next step

What changed after validation

- Removed fabricated island-level tourism pressure and GDP-intensity variables.
- Corrected NOAA Degree Heating Weeks parsing and preserved missing values as missing.
- Changed the target from same-row reconstruction to the next observed coral-cover change.
- Replaced random/group folds with expanding-year evaluation so training always precedes testing.
- Downgraded claims to screening-only because predictive gain is modest and causality is not identified.

Headline result

The selected Gradient Boosting model recorded MAE 5.90 percentage points per year versus 5.99 for the mean baseline—an improvement of 1.5%. Its forward-test R^2 was 0.051. This is enough for a cautious triage aid, not enough for automated enforcement.

1. Decision problem

Marine managers must decide where limited inspection, bleaching response, mooring checks, and waste-control resources should go first. National tourism totals are useful for context but cannot be credibly assigned to individual islands without island-level visitor-flow data. ReefSafe therefore focuses on the decision that the available evidence can support: which monitored islands require closer verification, and why?

1.1 Intended users and use

User	Decision	ReefSafe support
Marine-park officer	Schedule field checks	Priority rank, evidence, uncertainty
Reef scientist	Coordinate follow-up survey	Observed LCC, thermal stress, data confidence
Policy team	Allocate monitoring resources	National trend and transparent model benchmark

1.2 Guardrails

- No island-level tourism values are inferred from state totals.
- Feature importance is described as association, not cause.
- A high rank triggers verification; it is not a closure order or carrying-capacity quota.
- Thermal evidence never becomes a claim that visitors caused bleaching.

1.3 Observed national context

The unweighted mean live coral cover among surveyed island observations was 57.1% in 2012 and 39.8% in 2025. Survey composition changes across years, so this descriptive line should not be read as a fixed-panel national estimate.

2. Data and provenance

Source	Role	Treatment
Reef Check Malaysia annual reports	Island-year coral, substrate, fish, invertebrate and narrative indicators	Primary ecological evidence; confidence field retained
NOAA Coral Reef Watch	Sea-surface temperature anomaly and Degree Heating Weeks	Matched to regional stations; missing remains blank
OpenDOSM tourism and GDP series	National/state descriptive context	Not downscaled into model features
DOSM Datathon 2026 booklet	Submission and evidence rules	No simulated or fabricated observations

2.1 Data quality controls

- 404 processed island-year rows and 46 retained fields.
- NOAA rows use the published DHW field, not the adjacent threshold-exceedance field.
- Coordinates, marine-park labels, source confidence, and extraction source are retained for audit.
- First observations without a previous survey have no change-rate target and are excluded from transition modeling.

2.2 Known evidence limitations

Survey coverage is uneven across islands and years. Narrative impact flags indicate that a stressor was mentioned in a report; they are not continuous measurements of intensity. Regional NOAA stations do not replace in-water temperature loggers. Results should be refreshed whenever extraction corrections or new survey years become available.

3. Modeling method

Each training example uses conditions recorded for an island at one survey and predicts the annualized live-coral-cover change observed at that island's next survey. Rows are ordered within island before the target is created. Evaluation uses the latest five target years as separate test periods; every test year is trained only on earlier target years.

3.1 Compared models

Model	MAE (pp/yr)	RMSE (pp/yr)	R ²
Baseline mean	5.991	8.100	-0.019
Ridge regression	5.993	7.909	0.029
Gradient boosting	5.902	7.817	0.051
Random forest	6.123	8.073	-0.012

3.2 Model selection

Gradient Boosting had the lowest forward-test MAE (5.902 pp/yr). The difference from the baseline is only 1.5%; therefore the model is retained only for relative screening. The dashboard keeps the baseline comparison and R² visible so users cannot mistake the ranking for high-accuracy forecasting.

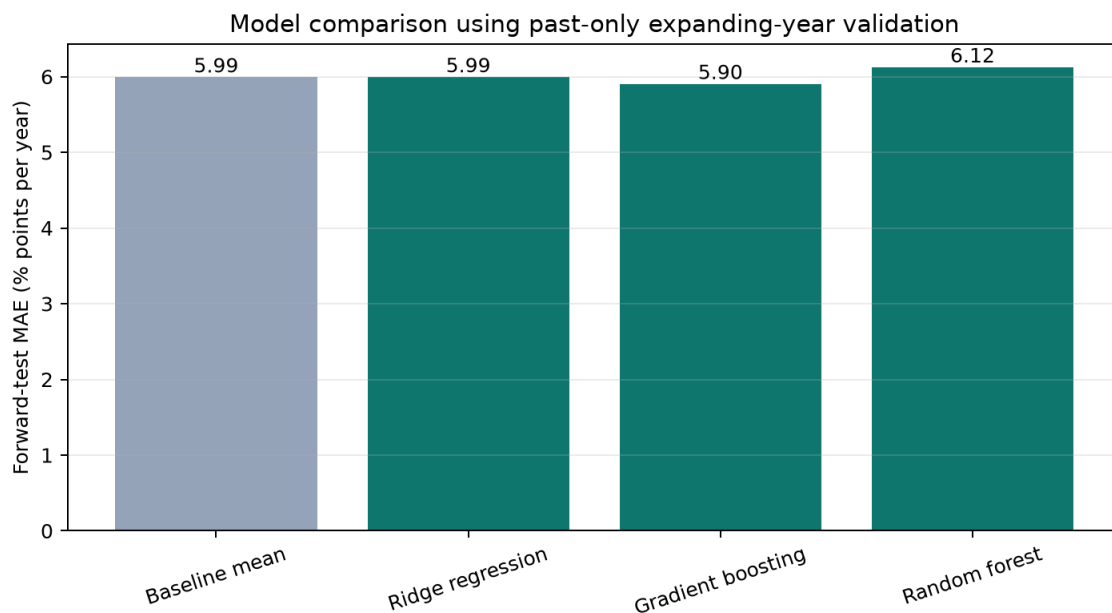


Figure 1. Forward-test model comparison. Lower MAE and RMSE are better.

4. Results and interpretation

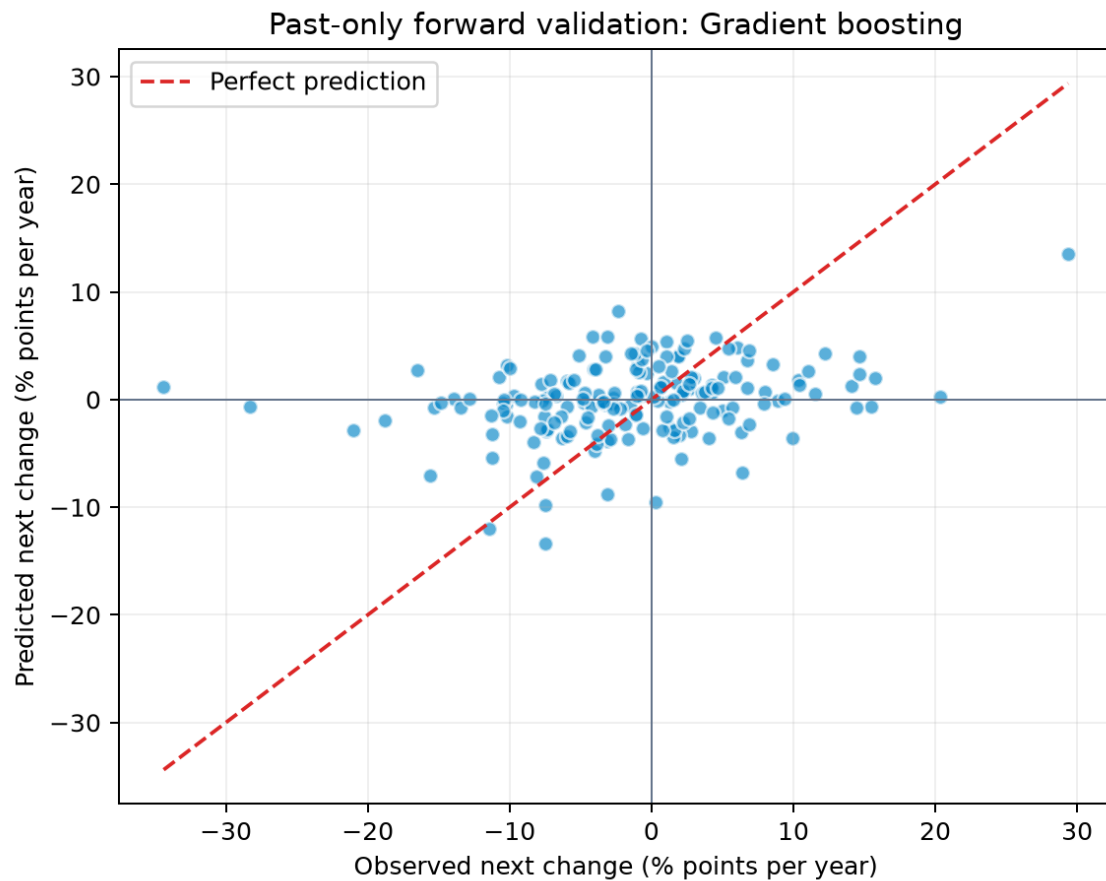


Figure 2. Held-forward observations versus predictions for the selected model.

Predictions remain concentrated around the centre of the observed distribution and do not reproduce extreme annual changes well. Wide empirical prediction bands are shown in the dashboard. This behaviour is consistent with the low positive R^2 and reinforces the need for field verification.

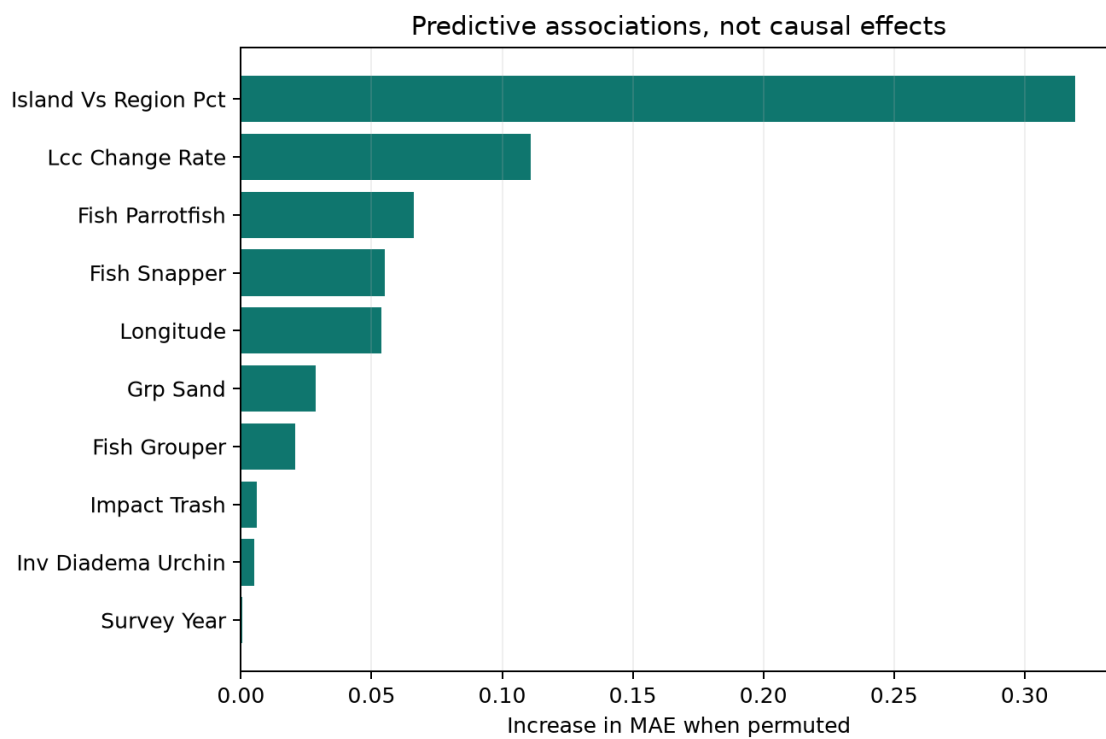


Figure 3. Forward-test permutation importance. Values indicate predictive association, not causation.

4.1 Factor relationships

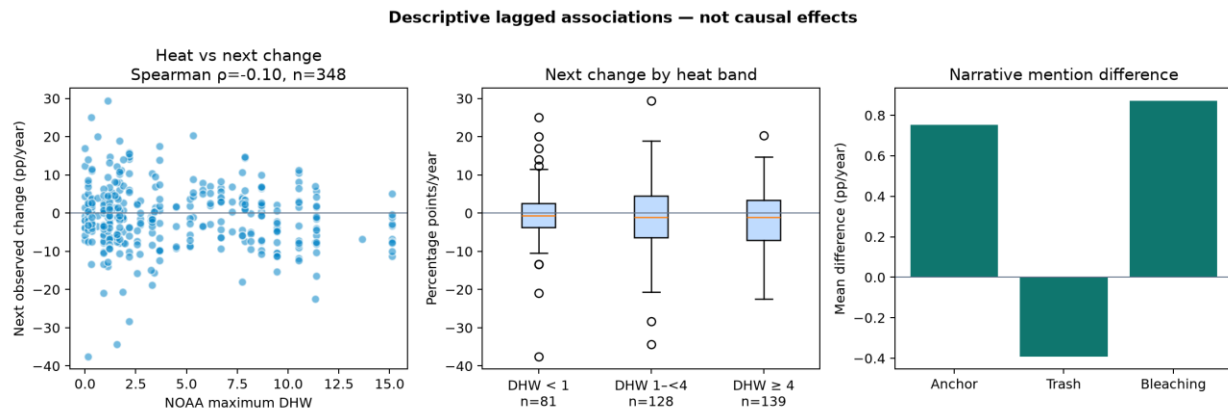


Figure 4. Factors measured at the current observation versus the next observed coral-cover change. These are descriptive lagged associations, not causal effects.

Maximum NOAA DHW had a weak negative Spearman relationship with the next observed coral-cover change ($\rho = -0.105$, $p = 0.051$, $n = 348$). The mean next change was -0.55 pp/year for DHW below 1 and -1.38 pp/year for DHW at least 4. This pattern is directionally consistent with heat stress, but the regional station match, irregular surveys, repeated islands and unmeasured local events prevent causal interpretation.

Narrative anchor, trash and bleaching flags did not show strong group separation in this sample. Absence of a narrative mention is not confirmed absence, so these indicators remain evidence prompts rather than causal or predictive conclusions.

5. Priority output

The latest available record for each island is scored by the final model. The top quartile is labelled ‘High screening priority’; all other islands remain in ‘Monitor’. Evidence and recommended actions are rule-based and intentionally conservative.

Rank	Island	State	Prediction	Observed evidence	Next step
1	Labuan	Sabah	-12.0	NOAA thermal stress observed	Coordinate bleaching survey; do not attribute thermal loss to visitors
2	Kapas	Terengganu	-4.5	Modelled decline; no single dominant observed stressor	Prioritise field validation before imposing restrictions
3	Lankayan	Sabah	-3.8	NOAA thermal stress observed	Coordinate bleaching survey; do not attribute thermal loss to visitors
4	Seri Buat	Pahang	-3.6	Anchor impact mentioned in Reef Check report	Inspect mooring availability and anchoring controls
5	Mertang	Johor	-3.4	Anchor impact mentioned in Reef Check report	Inspect mooring availability and anchoring controls
6	Mensirip	Johor	-3.2	Waste or pollution indicator is elevated	Inspect waste and wastewater controls
7	Tenggol	Terengganu	-3.1	Waste or pollution indicator is elevated	Inspect waste and wastewater controls
8	Sibu	Johor	-3.1	Anchor impact mentioned in Reef Check report	Inspect mooring availability and anchoring controls
9	Tinggi	Johor	-3.0	Anchor impact mentioned in Reef Check report	Inspect mooring availability and anchoring controls
10	Tun Sakaran Marine Park	Sabah	-3.0	NOAA thermal stress observed	Coordinate bleaching survey; do not attribute thermal loss to visitors

5.1 Evidence-to-action rules

- $DHW \geq 4$: coordinate bleaching assessment; do not attribute heat-driven loss to visitors.
- Anchor impact mentioned: inspect mooring availability and anchoring controls.
- Waste/pollution indicator elevated: inspect waste and wastewater controls.
- No dominant observed stressor: prioritise field validation before restrictions.

6. Dashboard and operating workflow

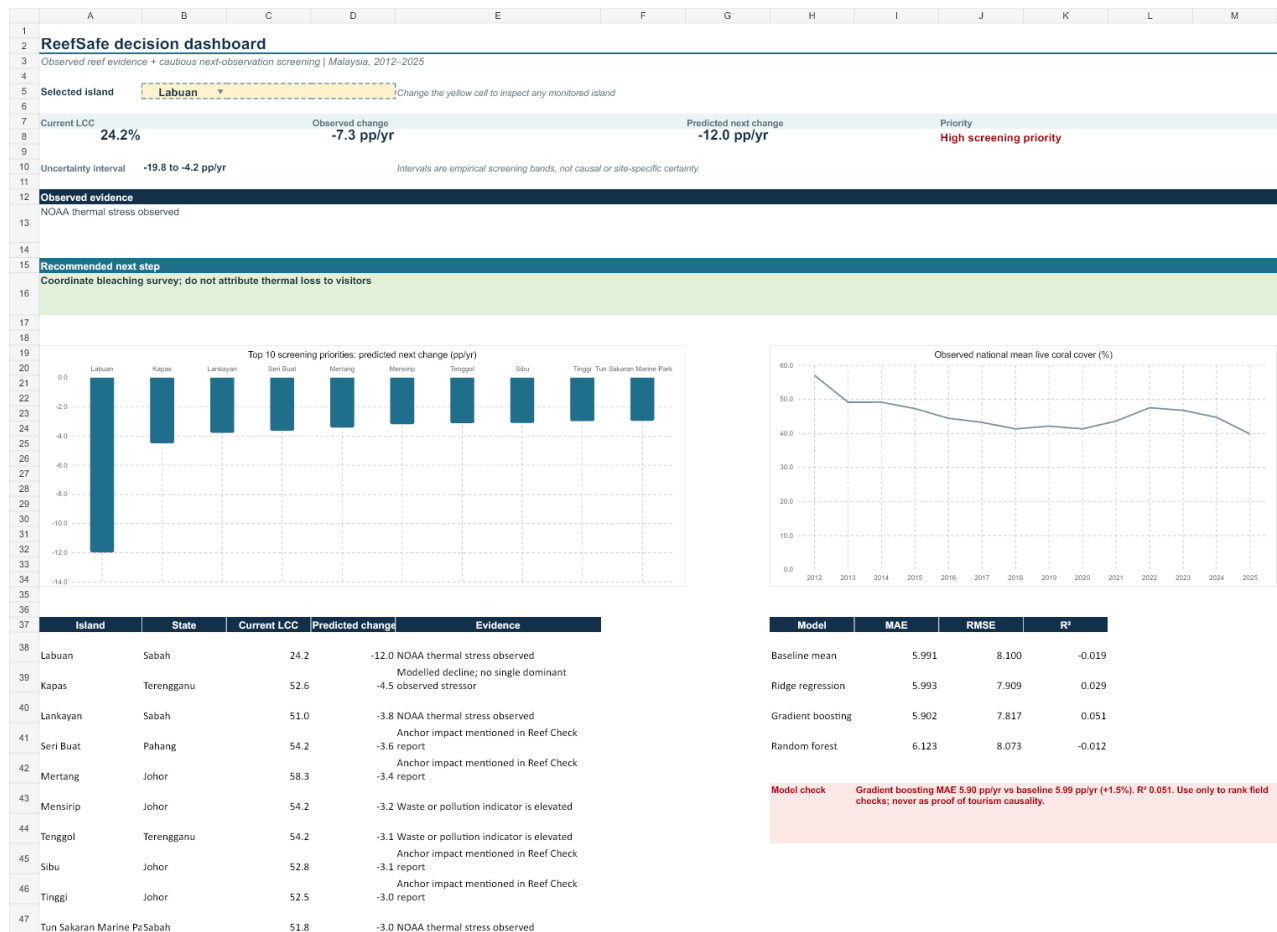


Figure 5. Interactive Excel dashboard. The yellow island selector drives formula-linked evidence and actions.

6.1 Recommended operating cycle

Step	Owner	Action
1. Refresh	Data analyst	Run preprocessing and model scripts after new reports arrive
2. Review	Reef scientist	Check source confidence, uncertainty, and observed stressors
3. Verify	Marine-park team	Conduct targeted field checks before enforcement
4. Record	Programme lead	Log action, outcome, and corrected evidence for the next cycle

7. Limitations, next data, and conclusion

7.1 Limitations

- The evaluation sample is small and temporally irregular; several island histories are sparse.
- The model's 1.5% MAE gain and R^2 0.051 are weak; rank stability should be monitored after each refresh.
- No island-level visitor count, vessel movement, wastewater load, enforcement, or revenue series is available in the modeled table.
- Prediction bands are empirical and should not be treated as formal confidence intervals for policy liability.

7.2 Project assumptions

Assumption	Risk	Mitigation
Island names represent comparable monitoring units	Site composition may change	Retain source confidence and verify large movements
Live coral cover is sufficiently comparable across reports	Method or observer changes may affect values	Use results for screening only
Annualized change is approximately linear between surveys	Intermediate disturbance and recovery are unseen	Keep survey and target years explicit
NOAA regional station represents broad heat exposure	Local reef temperature can differ	Request in-water confirmation
No narrative mention is uncertain absence	Reporting intensity varies	Treat flags as evidence prompts
The top quartile is a workload queue	It is not a scientific threshold	Require field verification

The full assumption register is maintained in docs/assumptions.md. OpenDOSM state and national totals are explicitly not assumed to measure individual-island tourism pressure.

7.3 Highest-value next data

The most useful enhancement is not a larger algorithm. It is island-level operational data: daily visitors, boat trips, mooring use, wastewater discharge, closures, bleaching observations, and consistent site-level survey identifiers. Those measurements would allow explicit pressure-response analysis and later testing of intervention effects.

7.4 Conclusion

ReefSafe turns heterogeneous public evidence into a transparent queue for field verification. Its professional value lies in traceability and restraint: the dashboard shows what is observed, what is predicted, how uncertain the prediction is, and what claim the data cannot support.

References and reproducibility

Primary evidence: Reef Check Malaysia annual survey reports; NOAA Coral Reef Watch station time series; OpenDOSM structured tourism/GDP files; DOSM Datathon 2026 booklet. Reproducible commands and packaged source files are listed in the repository README and dashboard-package README. Processed data: data/processed/master_reef_tourism_dataset.csv. Evaluation: data/processed/model_evaluation_metrics.csv. Diagnostics: data/processed/factor_relationships.csv and heat_category_summary.csv.